MSDM5004 Spring 2026

Exercise 1

Feb 23

Find one iteration using (1) the Jacobi method and (2) the Gauss-Seidel method for the following linear system, starting from $\mathbf{x}^{(0)} = (1, 2, 3)^T$.

$$\begin{aligned} 5x_1 + 2x_2 + x_3 =& -12, \\ -x_1 + 4x_2 + 2x_3 =& 20, \\ 2x_1 - 3x_2 + 10x_3 =& 3. \end{aligned}$$

Solution.

(1) The iterative scheme of the Jacobi method is

$$\begin{aligned}
x_1^{(k)} &= & & -\frac{2}{5}x_2^{(k-1)} & -\frac{1}{5}x_3^{(k-1)} & -\frac{12}{5}, \\
x_2^{(k)} &= & \frac{1}{4}x_1^{(k-1)} & & -\frac{1}{2}x_3^{(k-1)} & +5, \\
x_3^{(k)} &= & -\frac{2}{10}x_1^{(k-1)} & +\frac{3}{10}x_2^{(k-1)} & & +\frac{3}{10}.
\end{aligned}$$

From the initial condition $\mathbf{x}^{(0)} = (1, 2, 3)^T$, we have

$$\begin{aligned}
x_1^{(1)} &= -\frac{2}{5}x_2^{(0)} - \frac{1}{5}x_3^{(0)} - \frac{12}{5} = -\frac{2}{5}\cdot 2 - \frac{1}{5}\cdot 3 - \frac{12}{5} = -\frac{19}{5}. \\
x_2^{(1)} &= \frac{1}{4}x_1^{(0)} - \frac{1}{2}x_3^{(0)} + 5 = \frac{1}{4}\cdot 1 - \frac{1}{2}\cdot 3 + 5 = \frac{15}{4}. \\
x_3^{(1)} &= -\frac{2}{10}x_1^{(0)} + \frac{3}{10}x_2^{(0)} + \frac{3}{10} = -\frac{2}{10}\cdot 1 + \frac{3}{10}\cdot 2 + \frac{3}{10} = \frac{7}{10}.
\end{aligned}$$

(2) The iterative scheme of the Gauss-Seidel method is

$$\begin{aligned}
x_1^{(k)} &= & & -\frac{2}{5}x_2^{(k-1)} & -\frac{1}{5}x_3^{(k-1)} & -\frac{12}{5}, \\
x_2^{(k)} &= & \frac{1}{4}x_1^{(k)} & & -\frac{1}{2}x_3^{(k-1)} & +5, \\
x_3^{(k)} &= & -\frac{2}{10}x_1^{(k)} & +\frac{3}{10}x_2^{(k)} & & +\frac{3}{10}.
\end{aligned}$$

From the initial condition $\mathbf{x}^{(0)} = (1, 2, 3)^T$, we have

$$\begin{aligned}
x_1^{(1)} &= -\frac{2}{5}x_2^{(0)} - \frac{1}{5}x_3^{(0)} - \frac{12}{5} = -\frac{2}{5}\cdot 2 - \frac{1}{5}\cdot 3 - \frac{12}{5} = -\frac{19}{5}. \\
x_2^{(1)} &= \frac{1}{4}x_1^{(1)} - \frac{1}{2}x_3^{(0)} + 5 = \frac{1}{4}\cdot\left(-\frac{19}{5}\right) - \frac{1}{2}\cdot 3 + 5 = \frac{51}{20}. \\
x_3^{(1)} &= -\frac{2}{10}x_1^{(1)} + \frac{3}{10}x_2^{(1)} + \frac{3}{10} = -\frac{2}{10}\cdot\left(-\frac{19}{5}\right) + \frac{3}{10}\cdot\frac{51}{20} + \frac{3}{10} = \frac{73}{40}.
\end{aligned}$$